RD INFRO TECHNOLOGY

Machine Learning Internship Program

Join RD INFRO TECHNOLOGY's vibrant community. This program focuses on leadership development, learning, and fostering shared interests in machine learning.

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About RD INFRO TECHNOLOGY

Community Focus

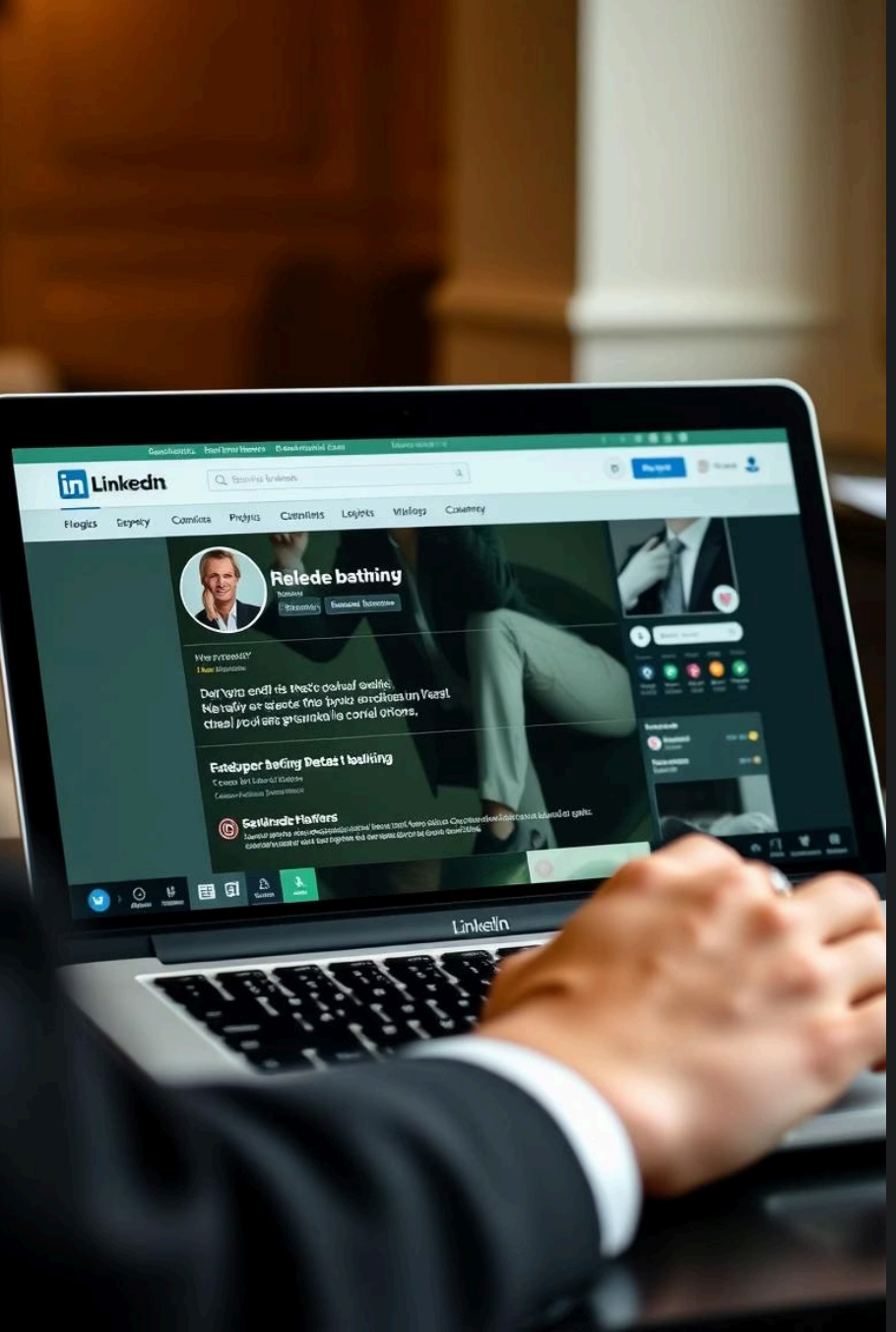
We are a diverse community uniting individuals with shared objectives and goals.

Opportunity Creation

Our main focus is creating opportunities in leadership, learning, and student engagement.

Leadership Development

We provide platforms and resources to develop leadership skills and drive positive change.



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Internship Requirements

1 LinkedIn Profile

Update your LinkedIn profile to reflect your participation.

2 Task Completion

Complete any one Task (1, 2, or 3) for successful internship completion.

3 GitHub Repository

Maintain a separate GitHub repository named "RD INFRO TECHNOLOGY" for all tasks.

Submission Guidelines

Submission Form

A task submission form will be shared later via email.

Create Video

Create a video to showcase your work and demo your effort.

Host on LinkedIn

Host the video on LinkedIn for proof and credibility.

Add Hashtags

Add #RDINFROTECH and #internship to your LinkedIn video posts.

Task 1: Problem Statement & Objective Definition

- **Description**: Define the goal (e.g., prediction, classification, clustering).
- **Tools**:
  - Google Docs, Jira, Miro (for brainstorming)
- **How to Use**:
  - Write business question → Convert to ML objective
    Example: “Will the customer churn?” → Binary classification

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Task 2: Data Collection

- **Description**: Gather data from files, databases, APIs, or web scraping.
- **Tools**:
- Python (Pandas, Requests, BeautifulSoup), SQL
- **How to Use**:

```
python

import pandas as pd   df = pd.read_csv('data.csv')
```

Task 3: Data Cleaning & Preprocessing

- **Description**: Remove nulls, encode categories, normalize values.
- **Tools**:
  - Pandas, Scikit-learn, OpenRefine
- **How to Use**:

```python
df.dropna(), df.fillna(), pd.get_dummies(df), StandardScaler()
```

Task 4: Exploratory Data Analysis (EDA)

- **Description**: Analyze patterns, correlations, and distributions.
- **Tools**:
- Matplotlib, Seaborn, Plotly
- **How to Use**:

```python
import seaborn as sns
sns.heatmap(df.corr(), annot=True)
```

Task 5: Feature Engineering & Selection

- **Description**: Create new features or reduce less useful ones.
- **Tools**:
  - Scikit-learn, FeatureTools
- **How to Use**:

```python
from sklearn.feature_selection import SelectKBest
X_new = SelectKBest(k=10).fit_transform(X, y)
```

Task 6: Model Training

- **Description**: Train machine learning models (e.g., Decision Tree, Random Forest, SVM).
- **Tools**:
  - Scikit-learn, XGBoost, TensorFlow, PyTorch
- **How to Use**:

```python
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier().fit(X_train, y_train)
```

Task 7: Model Evaluation & Tuning

- **Description**: Evaluate model performance and tune hyperparameters.
- **Tools**:
  - Scikit-learn (metrics, GridSearchCV), TensorBoard
- **How to Use**:

```python
from sklearn.metrics import classification_report  print(classification_report(y_test, model.predict(X_test)))
```

Task 8: Model Deployment

- **Description**: Convert the model into an API or UI tool.
- **Tools**:
  - Flask / FastAPI, Docker, Streamlit, AWS/GCP
- **How to Use**:
  - Save model:

```
python
```

```
import joblib  joblib.dump(model, 'model.pkl')
```

  - Create an API:

```
python
```

```
from flask import Flask, request  app = Flask(__name__)  @app.route('/predict', methods=['POST'])  def predict():      data = request.json      return model.predict([data])
```

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Support and Guidance

1

Purpose

The internship aims for your learning and growth.

2

Guidance

Seeking guidance is entirely your choice.

3

Approach

Approach tasks with professional diligence and attention.